

Discrete Curvature, Spin, and Geometric Algebra as a Modern Basis for Heim-Style Minimal-Area Physics

Motivations and constraints for a discrete, causal “deformation-only” model

A mathematically clean way to respect *locality* and *causality* in any discrete spacetime proposal is to ensure that the fundamental equations are formulated as **local update rules** (or a **local variational principle**) on a discrete structure (cell complex, lattice, triangulation), so that influences propagate through adjacency rather than by global reconfiguration. This is aligned with the standard continuum picture of relativistic field theories in which well-posed hyperbolic evolution (in appropriate gauges/formulations) yields a domain-of-dependence property rather than instantaneous action at a distance. ¹

In practice, discrete quantum gravity and discrete geometry communities already treat “geometry as the dynamical field” rather than as a fixed background, and they do so using strongly local building blocks (simplices, cells, links) together with an action/energy functional. For example, **Causal Dynamical Triangulations** is explicitly designed to preserve Lorentzian causal structure at the discrete level by using Lorentzian building blocks and gluing rules that enforce a well-behaved causal structure. ²

A second nonnegotiable, given your interest in a “minimal area quantum,” is that *area* (or closely related geometric data) should not be a derived afterthought. Here, it matters whether “minimal area” is (i) a **hard regulator** (lattice spacing/triangle areas fixed), (ii) a **quantized spectrum** of an area observable, or (iii) an **emergent effective discreteness** from dynamics. Loop-quantum-gravity-style constructions famously produce discrete spectra for geometric operators (area and volume), providing an existence proof that “minimal area” can arise as a spectrum rather than as an imposed cutoff. ³

These points set up a concrete target: a discrete, local theory with a well-defined energy/action tied to curvature and/or connection holonomy, in which “particles” can be interpreted as localized, propagating field configurations (defects/solitons/excitations) rather than as fundamental point objects. Classical field theory already contains robust examples of stable, particle-like field configurations (topological solitons), which is conceptually compatible with your “there are no particles; only moving deformation” intuition. ⁴

What Heim actually claims about discreteness and six dimensions

In the most directly accessible English-language source text attributed to Heim, [entity]“people”, “Burkhard Heim”, “German physicist”] motivates discreteness via action quantization and then introduces a **smallest geometrical unit** that acts as a cutoff for infinitesimal analysis, i.e., a statement that the differential calculus of arbitrarily small quantities is not physically applicable below that unit. ⁵

He also explicitly argues for a **six-dimensional reference space** R^6 containing ordinary spacetime R^4 as a subspace, and introduces extra coordinates (described there as imaginary coordinates x_5, x_6) to accommodate the structure of his energy-density tensor and its components. ⁶

Most importantly for your “minimal area quantum” direction, Heim’s 1976 presentation asserts a natural constant τ that he describes as a **minimum geometrical element** of dimension area (an “elementary surface”), and he gives it a numerical value, commenting on its relation (up to a factor) to the square of the Planck length. ⁷

Finally, in the same body of material, Heim characterizes “particles” in geometric/structural terms (e.g., as an “elementary flow system”) rather than as fundamental classical points, and he sketches the use of discrete pre-structures (e.g., “protosimplexes”) in building up his intended framework. ⁸

These claims are clearly *structurally adjacent* to modern discrete geometry and gauge-gravity formalisms (because they emphasize local building blocks and a smallest geometric scale), but Heim’s work—as you note—does not, in these materials, cleanly present a widely adopted, modern action functional that makes the curvature/metric-energy relationship explicit in the way contemporary discrete GR does. ⁹

Six “dimensions” as bivectors: a plausible reinterpretation inside Clifford and Cartan formalisms

A central technical observation is that the number **six** arises *very naturally* in 4D geometry without requiring six spacetime coordinates.

From the perspective of Clifford/geometric algebra, the algebra is graded: its grade- k subspace consists of k -vectors. In standard constructions, the full algebra over an n -dimensional vector space has dimension 2^n , and the grade- k piece is spanned by basis elements corresponding combinatorially to subsets of size k . ¹⁰ As a result, in $n = 4$, the bivector (grade-2) subspace has dimension $\binom{4}{2} = 6$. ¹⁰

Independently (and compatibly), Lie theory tells you that the rotation group in n dimensions has dimension $n(n - 1)/2$, and the Lorentz group’s Lie algebra likewise has six generators in 4D because it is built from antisymmetric index pairs. ¹¹

This becomes conceptually concrete in spacetime algebra treatments: the **bivector basis** forms the Lie algebra of the Lorentz group under the commutator bracket, and “the Lorentz group generators are the 6 bivectors,” with three corresponding to spatial rotations and three to boosts; exponentiating bivectors produces Lorentz transformations. ¹²

Now connect this to modern gravity formalisms:

In **Cartan / tetrad + spin-connection** formulations of gravity, curvature is naturally a **bivector-valued** object (a Lorentz-algebra-valued 2-form). A standard geometric interpretation in the torsion/curvature literature is that torsion is tied to translations and curvature to rotations, and that torsion and curvature can be viewed as 2-form “misfits” measured by a Cartan circuit; the curvature is explicitly described as bivector-valued because it corresponds to infinitesimal rotations. ¹³

This gives a very plausible failure mode for “extra dimensions” in older or idiosyncratic notational systems: what is *really* a **six-component internal bivector index** (the Lorentz algebra’s antisymmetric pair ab) can be misread as a **six-dimensional coordinate** label. In modern language, those six degrees of freedom live in the tangent-space rotation algebra (or in the grade-2 subspace of a Clifford algebra), not in an additional spacetime manifold dimension. ¹⁴

This does not prove Heim “meant bivectors,” because Heim’s own text explicitly introduces R^6 with extra coordinates x_5, x_6 . ⁶

But it does establish a mathematically standard alternative interpretation of “six” that is both (i) well established and (ii) directly relevant to curvature, holonomy, and spin—exactly the ingredients you want to connect to an energy functional. ¹⁵

Discrete curvature with locality: triangulations, DEC, and holonomy

Modern discrete differential geometry gives at least three complementary “locality-friendly” ways to represent curvature.

Discrete differential forms and operators on meshes

Discrete Exterior Calculus (DEC) is an explicit attempt to build a calculus “on a discrete space,” defining discrete differential forms, discrete vector fields, and discrete operators (exterior derivative, Hodge star, Lie derivative, etc.) on simplicial complexes, typically using a primal complex and a dual complex (often circumcentric duals). ¹⁶

This supports the kind of “chemistry-like exchange” intuition you mentioned because DEC-like constructions naturally represent fluxes, circulation, and local conservation constraints discretely. ¹⁶

Curvature from angle defects: local, concentrated curvature

At the level of discrete surfaces/meshes, curvature can be represented by local angle defects. In lecture notes designed for computation, `Identity["people","Keenan Crane","computer graphics researcher"]` emphasizes how discrete analogs can preserve geometric structure and how angle sums around a vertex capture deviations from flatness (and how one can conceptualize “pushing curvature into otherwise flat triangles”). ¹⁷

This is a discrete geometric version of “local change of angles,” and it is explicitly local: the curvature information lives at (or near) combinatorial features like vertices/hinges. ¹⁸

Regge calculus: curvature on codimension-2 hinges and a discrete gravitational action

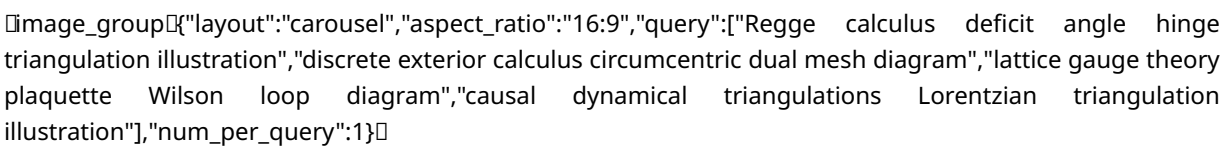
In 4D Regge calculus, curvature is concentrated on triangles (“hinges,” codimension-2 simplices). A modern summary states explicitly that Regge’s 1961 work approximates curved spacetimes by simplicial complexes with curvature distributed on codimension-2 simplicial sets. ¹⁹

A widely used expression is that the Einstein–Hilbert action is approximated by a sum over hinges of (area) $\times$ (deficit angle), and recent variational discretizations build on this. ²⁰

Holonomy and lattice gauge theory viewpoint

Your emphasis on “moving deformation,” local angle changes, and causality maps extremely well onto **holonomy-based** discretizations. In lattice gauge theory, the fundamental variables are group-valued link holonomies (parallel transport), and gauge-invariant curvature information is captured by plaquette holonomies (Wilson loops); the Wilson action is built from these plaquette variables and reproduces the continuum Yang–Mills action at leading order. ²¹

This holonomy viewpoint has already been applied to gravity in first-order formalisms: one discrete gravity construction in Palatini spirit defines connection matrices between simplex frames, uses holonomy around hinges, and has an action explicitly described as “very similar to the Wilson action” while being locally invariant under the Poincaré group. ²²



Energy functionals that tie curvature, metric, and spin, with discrete realizations

The missing step you identified in Heim—an explicit energy/action functional tied to curvature/metric and (ideally) to spin—has several well-established modern candidates. Importantly, these candidates already come with known discretizations that preserve locality.

Curvature energy from the Einstein–Hilbert action via Regge calculus

A canonical curvature energy in GR is the Einstein–Hilbert action, and Regge calculus provides a standard discretization by replacing the integral of scalar curvature with a hinge sum. A modern discrete-gravity paper states this explicitly in 4D: Regge calculus approximates the Einstein–Hilbert action by $\sum_h A_h \delta_h$, where δ_h is a deficit angle measuring the failure of simplices around a hinge to fit flatly. ²³

If you want an “energy function that relates to curvature,” this is the cleanest mainstream answer: it is literally a discrete scalar-curvature action, and it supports local modifications of geometry by local changes to simplex data. ²⁴

First-order (tetrad + connection) action: Palatini / Einstein–Cartan–Sciama–Kibble form

If your foundational intuition prioritizes *spin* and “geometric exchange” more than a metric-first approach, then the first-order formalisms are natural. In a tetradic Palatini formulation, the Lagrangian involves the tetrad e , spin connection ω , and curvature R , and the action is sometimes discussed in relation to the Einstein–Cartan–Sciama–Kibble (ECSK) form. ²⁵

Discrete realizations exist that are explicitly built using lattice gauge theory ideas and Cartan geometry, producing algorithms that preserve local Lorentz gauge symmetry and have a variational structure. ²³

This matters for your “six dimensions might be bivectors” suspicion because, in this language, the spin connection is a **Lorentz-algebra-valued** 1-form (six generators in 4D), and curvature is correspondingly Lorentz-algebra-valued (bivector-valued). ¹⁵

Spin → torsion coupling and “contact” interactions

If the “basic level” is supposed to encode something like spin-induced exchange without forces in the usual sense, torsion-based gravity theories provide a rigorous framework where spin couples to geometry differently than mass-energy does.

A review-article abstract in the GR/torsion literature states that allowing torsion leads to a theory that unifies gravitation with a very weak **spin-spin contact interaction**. ²⁶

A separate, more geometric essay explains how torsion is related to translations (as curvature is related to rotations) and explicitly ties torsion to dislocation density in a “continuized crystal,” making the defects/deformations analogy mathematically precise. ²⁷

If your ontology is “moving deformation” rather than “moving particles,” this defect/translation-rotation picture is unusually compatible: excitations can literally be treated as propagating geometric defects (torsion/curvature concentrations) rather than as point particles. ²⁸

Gauge-holonomy energy: Wilson-type actions as “curvature penalties”

If you prefer to avoid scalar curvature at the start and build everything from local parallel transport, then the lattice gauge view is: energy is a sum over plaquettes of a function of holonomy (Wilson action), whose continuum limit yields a curvature-squared term (Yang–Mills). ²¹

This is mathematically attractive for “no instantaneous long-range updates” because the plaquette holonomy is *intrinsically local*—it depends only on a small loop. It also interfaces naturally with Clifford/geometric algebra because bivectors are exactly the objects that encode infinitesimal rotations/boosts (i.e., the Lie algebra elements you exponentiate to get group elements). ²⁹

A synthesis that best matches your Heim-motivated direction

The strongest “modern, well-established” synthesis that hits all your stated requirements is:

Use a **discrete Cartan / gauge-gravity formulation** on a simplicial (or more general cell) complex, with **group-valued holonomies** (Spin/Lorentz/Poincaré, depending on your choice) on links/dual edges, and an **action built from local holonomy data** (and possibly also tetrads/coframes), optionally constrained or quantized to enforce a minimal area scale.

This is not speculative: (i) first-order discrete gravity explicitly built from Poincaré/Lorentz gauge ideas exists, (ii) such constructions are explicitly related to Wilson-like lattice gauge actions, and (iii) Regge’s curvature-by-deficit-angle picture gives a direct discrete Einstein–Hilbert curvature energy. ³⁰

To incorporate “minimal area,” you have two clean modern options:

- **Hard cutoff option:** treat the triangulation’s building blocks as having a shortest scale (lattice spacing), as done in lattice approaches including CDT (where the lattice spacing is a regulator and the gluing rules enforce Lorentzian causal structure). ²
- **Quantized observable option:** treat areas as quantum observables with discrete spectra (loop-style), which conceptually aligns with Heim’s insistence on a smallest geometric unit while avoiding the need to add extra coordinates merely to “fit” components. ³¹

Finally, to align with “no particles,” you can adopt a field/defect ontology: localized, stable excitations can be treated as soliton-like or defect-like configurations, which is already standard in nonlinear field theory and is consistent with the torsion/dislocation analogy. ³²

Why this synthesis is the best match to your “six dimensions might be bivectors” suspicion

If Heim’s “ R^6 ” is re-read as “six bivector degrees of freedom” (Lorentz algebra / grade-2 subspace), then the modern Cartan + holonomy picture is almost exactly the mathematically standard implementation: curvature is bivector-valued (rotation-valued), torsion is translation-valued, and both are measurable via local circuits/holonomies. ³³

This removes the need to posit additional spacetime coordinates *in order to get six*, while still keeping *six* as a fundamental structural count. It also gives you a principled place to write down the energy functional Heim seems to omit in the materials you cited: you can choose Regge’s $\sum A\delta$, Palatini’s $e \wedge e \wedge R$, a Wilson-holonomy action, or hybrids, all of which are local and curvature-linked. ³⁴

A concrete “starting point” formalism that stays close to established mathematics

If you want a single recommended starting point that is both mainstream and maximally compatible with your goals, it is:

Discrete tetrad + spin-connection gravity (Palatini-style) implemented with holonomies on a simplicial/cellular complex, using a variational principle that reduces to Einstein–Hilbert/Regge in the appropriate limit, and that can be extended toward torsion/spin coupling if desired. ³⁵

This gives you:

- clear geometric primitives (cells, faces, hinges),
- a local curvature measure (holonomy/deficit),
- an explicit curvature-linked action,
- a natural home for spin/torsion,
- a mathematically standard interpretation of “six” as bivector/Lorentz-generator degrees of freedom,
- and an immediate pathway to “minimal area” either as regulator (CDT/lattice spacing) or as quantized spectrum (LQG-style area operator). ³⁶

1 <https://d-nb.info/1113636211/34>

<https://d-nb.info/1113636211/34>

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